Experiment Number: 5

Experiment Name: Transistors and Transistor Circuits

Station Number: 1

Submission Date: October 7th, 2025

## Objective

The objective of this experiment was to study the DC characteristics of transistors as well as their current-voltage relationships.

## Equipment

- (1) Computer with PSPICE software

## Procedure

***Step 1 – Observing Current Relationships***

1. Open the PSICE software by searching for "Capture" in the Windows menu and make a new file.
2. Create the circuit of figure 1 below by dropping parts, connecting with wires at nodes, and assigning values:

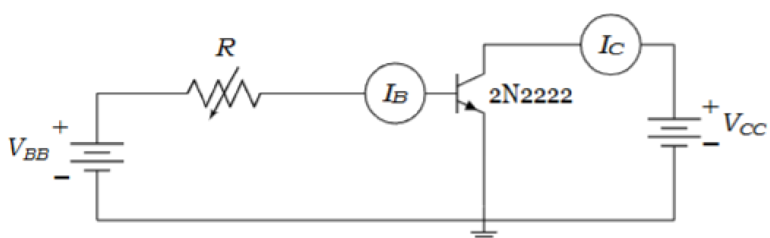


*Figure 1: The following transistor circuit will be used to observe current relationships along the transistor terminals.*

3. Assign Vbb as 2 V and Vcc as 5 V.
4. Add a parameters element by: opening the parts list (p) >> selecting "Special Library" >> searching for element "PARAM." Drop it anywhere on the screen.
5. Assign the value of the resistor R as: {Rval}.
6. Double click the PARAMETERS element to open the properties.

7. Assign the name and value of the PARAMETERS to the variable resistor by typing "Rval" with a default of 1kΩ. You have now declared the resistor as a global parameter.
8. Make a new simulation profile. Change the type from "Transient" to "DC Sweep."
9. Select "Global Parameters" and type the name of the global parameter as declared previously: "Rval"
10. Set the start and points to 1Ω and 100kΩ respectively with increments of 100.
11. Run the simulation and apply current markers and differential voltage markers to view the output signal as well as the base and collector currents.
12. On the waveform screen, select plot >> add traces >> input "Ic/Ib."
13. Observe and record the waveform. Note what the relationship entails.
14. Add a new trace, this time by selecting Ic only for the Y axis. Under "edit," change the x-variable from resistance to Ib.
15. Observe and record the waveform. Note what the relationship entails.

***Step 2 – Observing Voltage Relationships***

1. Using the same techniques as before, construct the circuit of figure 2a below. *Note: The missing resistance value is 12k Ω:

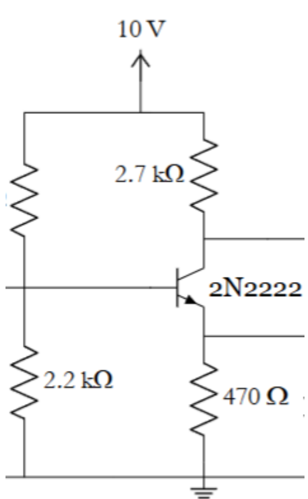

*Figure 2a: The following transistor circuit will be used to observe the voltage relationship between the input and output.*

2. Make a new simulation profile (running in "transient" this time.)
3. Apply differential voltage markers.
4. Run the simulation and observe the voltage drops at different points along the circuit.
5. Add to the circuit made to resemble that of figure 2b below:

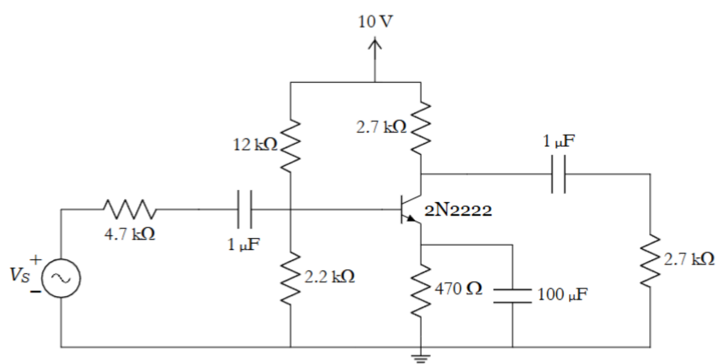


*Figure 2b: The following is the common emitter amplifier circuit used to observe the gain properties of a transistor amplifier circuit.*

6. Make a new "transient" simulation profile and apply differential voltage markers across the input and output.
7. Run the simulation and observe the gain in the waveform.
8. Record results and note the gain.

## Results

### ***Part 1 Results***

The following was recorded for this part of the procedure:

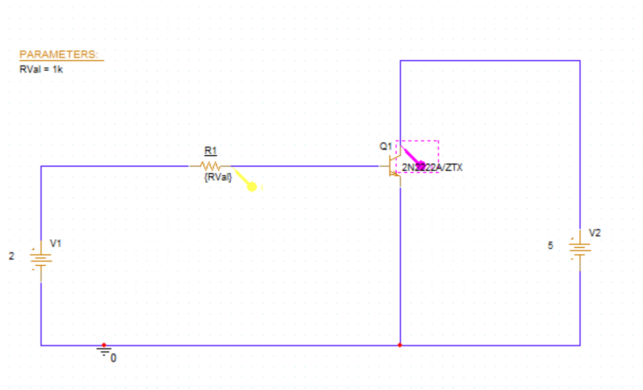


*Figure 3: The following is the simulated construction of the circuit of figure 1.*

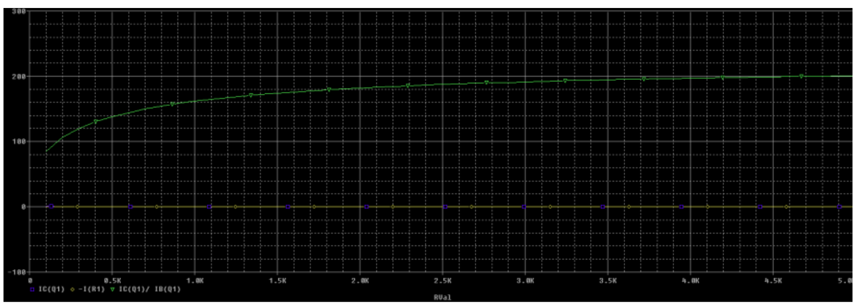


*Figure 3a: The following is the waveform of figure 3 when displaying the relationship between β and Rl.*

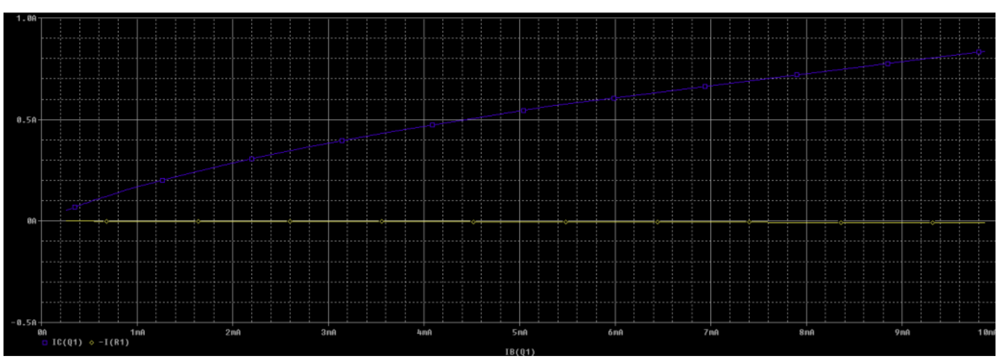


*Figure 3b: The following is the waveform of figure 3 when displaying the relationship between Ic and Ib.*

***Part 2 Results***

The following was recorded for this part of the procedure:

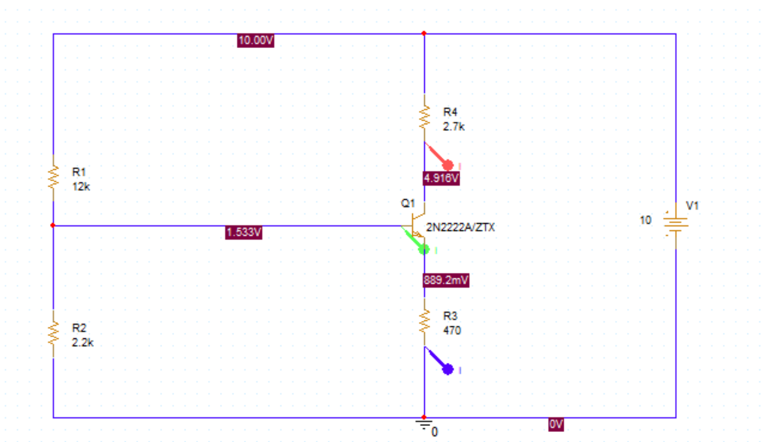


*Figure 4: The following is the simulated construction of the circuit of figure 2a.*

The voltage drop across the collector was recorded as 4.916 V, and the voltage drop across the base was recorded as 1.533 V.

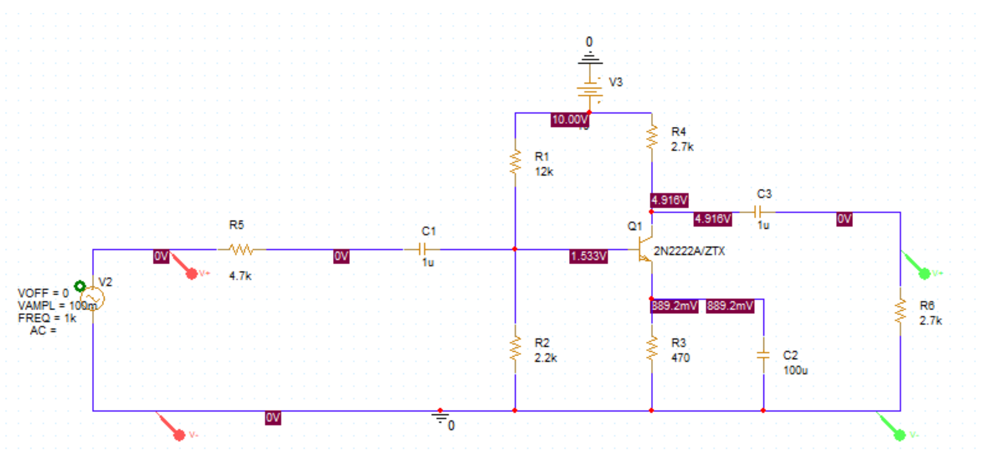


*Figure 5: The following is the simulated circuit of figure 2b.*

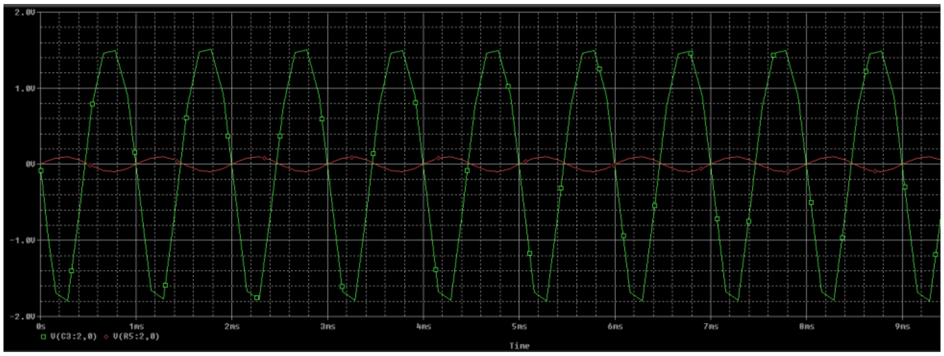


*Figure 6: The following is the waveform of figure 5 where the input signal is red and the output signal is green.*

We noted an increase from the signal voltage amplitude of approximately 0.1 V to 1.8 V at the output resulting in an **approximate gain of 18.**

**Report Questions**

No post lab questions were provided for this experiment. All measurements were required to be recorded only.

## Conclusion

In the first part of the experiment, we recalled the fact that β is the ratio of the collector current to the collector current. When studying the relationship of β and resistance we noted that as resistance increased, β increased as well until reaching a plateau point. In this sense, the ratio appeared logarithmically. It resembled the general saturation curve we studied in the lecture portion of this class. Comparing base current with collector current showed a similar effect in which an increase in base current resulted in an increase of collector current until a certain point.

The second portion of this lab is where we applied the concepts of transistor amplifiers. The first half was a simplified portion of the circuit which was used to show the amplifying effect they can have as the collector voltage was about 3.3 times greater than the base voltage. When connecting the rest of the circuit and applying a small AC signal as the input instead, we achieved a visual representation of the effect on AC signals as the peaks of the output voltage was visibly greater than the input. This was the most intuitive form of representing gain we have learned thus far. It is also worth noting the fact that the output signal was inverted in a sense which through us off at first, but upon further research, we confirmed that transistor amplifiers do indeed have this effect.

## Preparation Problems

1. **We will design the self-biased JFET circuit such that the current of the Drain is much less such that the JFET's saturating current (i.e. ID >> IDSS). How much less qualifies as "much less"**

   Much less in this case is when Id is less than 10% of Idds.

2. **When is ID = IDSS?**

   This is when Vgs is 0.

3. **What is the purpose of the capacitor that is in parallel with the 470 ohm resistor in the "Common-emitter amplifier" circuit (Figure 7)**

   This is to convert the current to a voltage signal simply using Ohm's law.

4. **What will be the polarity of the output signal with respect to the input signal for "Common-emitter amplifier" circuit (Figure 7)?**

   It will be inverted just like an inverting op-amp.

## Worksheet

No physical calculations were needed for this experiment.